



CPMAI Fundamentals

CPMAI (Cognitive Project Management for AI) provides a structured, repeatable framework for managing AI and data projects effectively. It integrates **PMI's Project Management principles** with **CRISP-DM (Cross Industry Standard Process for Data Mining)** to ensure business alignment and technical rigor.

Core Principles

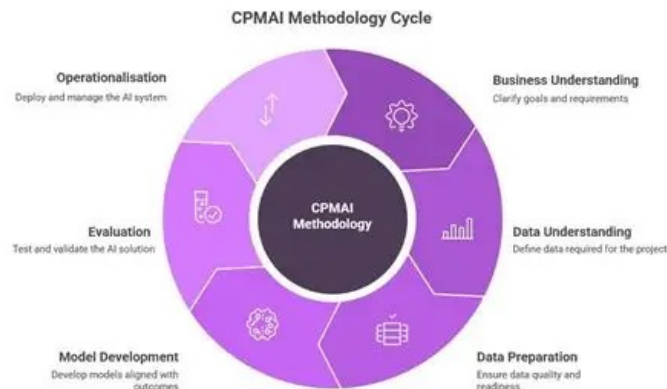
- Business-driven AI** : Every AI initiative must solve a real business problem.
- Iterative development** : Continuous refinement through feedback loops.
- Data-centric approach** : Data quality and understanding drive success.
- Governance and ethics** : Responsible AI practices are embedded throughout.



CPMAI Methodology Overview

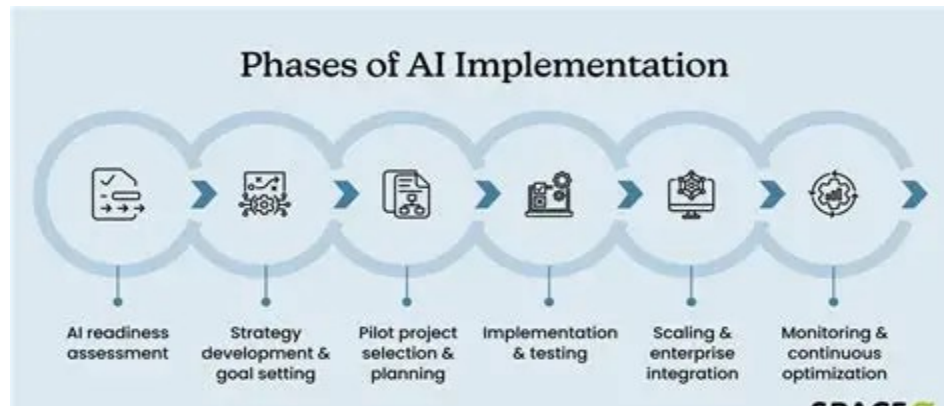
The methodology consists of **six iterative phases**, each building on the previous one:

<u>Phase</u>	<u>Focus Area</u>	<u>Key Activities</u>
1. Business Understanding	Define objectives and success criteria	Identify business problems, stakeholders, and AI feasibility
2. Data Understanding	Explore and assess data	Collect, describe, and evaluate data quality
3. Data Preparation	Clean and structure data	Data cleansing, labeling, transformation, and feature engineering
4. Model Development	Build and train models	Select algorithms, train, and tune models
5. Model Evaluation	Validate performance	Test against metrics and business goals
6. Model Operationalization	Deploy and monitor	Integrate models into production and ensure governance





The six iterative phases of CPMAI:



Phase I: Business Understanding

The phase where the project's purpose, objectives, and feasibility are defined. It ensures that AI is applied to solve a real business problem rather than being used for technology's sake.

Goal: Define the business problem and success criteria.

Activities:

- Identify stakeholders and objectives.
- Ask: *Do we need an AI approach to solve this problem?*
- Establish KPIs and measurable outcomes.

Output: Clear business case and feasibility assessment.



Phase II: Data Understanding

The phase focused on exploring and assessing available data to determine its relevance, quality, and suitability for solving the business problem.

Goal: Explore and assess the data available.

Activities:

- Collect datasets from internal and external sources.
- Profile data quality (completeness, consistency, bias).
- Identify gaps and risks.

Output: Documented data inventory and initial insights.



Phase III: Data Preparation

The phase where raw data is cleaned, transformed, and structured into a usable format for AI modeling. This is often the most time-consuming step.

Goal: Clean, transform, and structure data for modeling.

Activities:

- **Data cleansing:** Fix errors, handle missing values.
- **Data labeling:** Annotate for supervised learning.
- **Feature engineering:** Create meaningful variables.
- **Data integration:** Merge multiple sources.

Output: High-quality, structured dataset ready for modeling.



Phase IV: Model Development

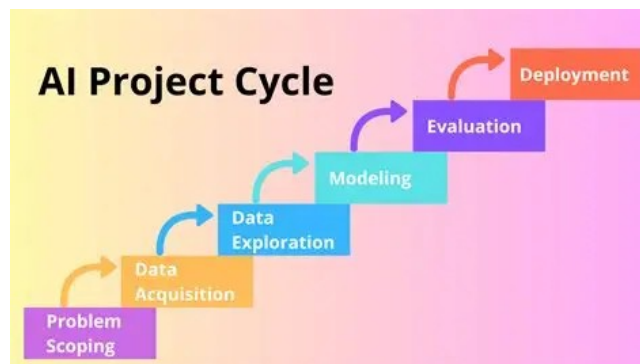
The phase where AI models are built, trained, and tuned using prepared data. It focuses on algorithm selection and prototype creation.

Goal: Build and train AI models.

Activities:

- Select algorithms appropriate for the problem.
- Train models using prepared data.
- Tune hyperparameters for optimization.
- Develop prototypes and test workflows.

Output: Candidate models with performance metrics.





The phase where models are validated against both technical performance metrics and business objectives to ensure they meet requirements.

Activities:

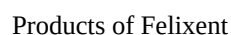
- Output:** Decision on whether the model is fit for deployment.



The phase where AI models are deployed into production, monitored continuously, and governed for compliance and ethics.

Activities:

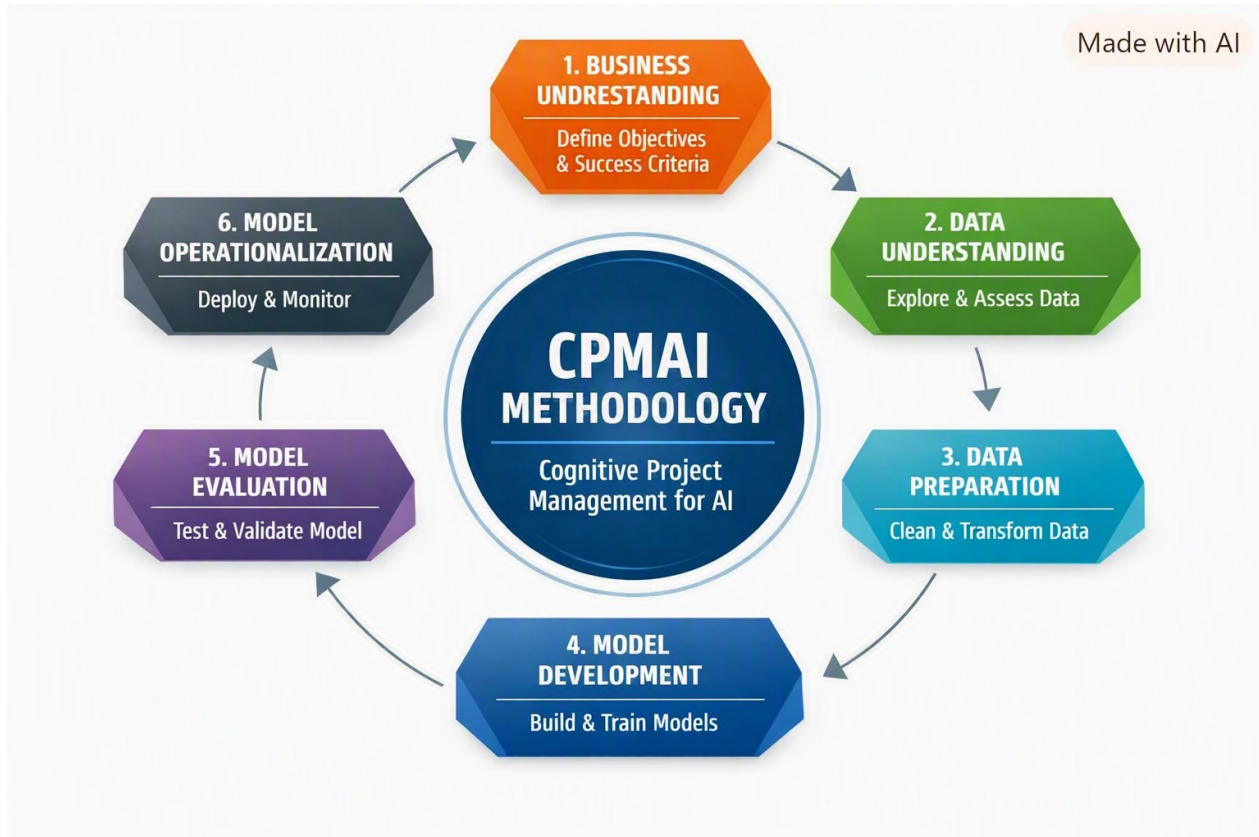
- Output:** Operational AI solution with monitoring and improvement plan.





Iterative Nature of CPMAI

The **CPMAI cycle is iterative** — after deployment, monitoring often reveals new data needs or business changes, which loop back into **Data Understanding** and **Data Preparation**. This ensures AI solutions evolve with the business.



Visual flow diagram

— the six iterative phases of CPMAI represented in a clean, circular sequence.

✓ The diagram shows;

Business Understanding → Data Understanding → Data Preparation → Model Development → Model Evaluation → Model Operationalization,

all connected in a loop to emphasize the **iterative nature** of the methodology.